Hades Model Report

v0.8.4 GitHub

Hastie & Smith, et al. (2026)

MODEL CONFIGURATION

This report summarises the calibration of the model against the reference dataset, including parameter estimates and convergence diagnostics.

Mode	**Subalkaline**
Submode	**Tholeiitic**
Data suite	**./synthetics.csv**
FeO/Fe^{2+}	**0.9**
Stages Run	Primary Primitive Basic Intermediate

1. Results at a Glance

All of the mineral abundances and fractional crystallisation (FC) values for each phase are shown in the table below, as well as the resultant total fractions for the performed amount of fractional crystallisation over all phases.

The total melt fraction crystallised across all phases is 64.30 wt.%

	Primary	Primitive	Basic	Intermediate	Total
Clinopyroxene	100.0	25.0	10.0	2.5	33.18
Orthopyroxene		75.0	45.0	40.0	41.24
Plagioclase			45.0	40.0	21.42
Amphibole				17.5	4.16
FC%	15.0	20.0	25.0	30.0	

Table 1: Optimised mineral weights for the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

2. Parameter Configuration

The parameters below define the model as it was executed for this run.

	Primary	Primitive	Basic	Intermediate	Evolved
Plagioclase	1.2	1.2	1.2	1.2	1.2
Clinopyroxene	0.28	0.28	0.28	0.28	0.28
Orthopyroxene	0.29	0.29	0.29	0.29	0.29
Amphibole	0.37	0.37	0.37	0.37	0.37
K-feldspar	1.3	1.3	1.3	1.3	1.3
Biotite	1.0	1.0	1.0	1.0	1.0
Titanaugite	1.0	1.0	1.0	1.0	1.0
Nepheline	1.0	1.0	1.0	1.0	1.0
Leucite	1.0	1.0	1.0	1.0	1.0
Nosean/Hauyne	1.0	1.0	1.0	1.0	1.0
Alkali-amphibole	1.0	1.0	1.0	1.0	1.0

Table 2: Partition coefficients for the model phases.

3. Magma Evolution

Initial Compositions

	Sample	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
Primary	Primary	51.0	1.0	15.0	9.0	12.5	10.0	1.0	0.5
Primitive	Primitive	52.94	1.13	17.47	7.59	7.65	11.54	1.13	0.54
Basic	Basic	54.68	1.21	20.65	6.55	4.43	10.7	1.21	0.57
Intermediate	Intermediate	56.87	1.46	21.95	6.66	2.48	8.89	1.08	0.6
Evolved	Evolved	64.01	1.53	25.0	0.5	0.71	6.61	0.94	0.69
End									

Table 3: Major oxide compositions of the model phases (wt.%).

Computed Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
Primary	51.0	1.0	15.0	9.0	12.5	10.0	1.0	0.5
Primitive	52.94	1.13	17.47	7.59	7.65	11.54	1.13	0.54
Basic	54.68	1.21	20.65	6.55	4.43	10.7	1.21	0.57
Intermediate	56.87	1.46	21.95	6.66	2.48	8.89	1.08	0.6
Evolved	64.01	1.53	25.0	0.5	0.71	6.61	0.94	0.69
End								

Table 4: Major oxide compositions of the model phases (wt.%).

4. Optimisation Results

Primary > Primitive

PRIMARY

Population size	100
Iterations	128/5000
Cost function	0.00
Trace analysis?	False

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	51.0	52.94	52.94	0.0
TiO2	1.0	1.13	1.13	0.0
Al2O3	15.0	17.47	17.47	0.0
Fe2O3	9.0	7.59	7.59	0.0
MgO	12.5	7.65	7.65	0.0
CaO	10.0	11.54	11.54	0.0
Na2O	1.0	1.13	1.13	0.0
K2O	0.5	0.54	0.54	0.0

Table 5: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	15.0
Olivine	0.0
Spinel	0.0
Clinopyroxene	100.0
Orthopyroxene	0.0
Plagioclase	0.0
Amphibole	0.0
K-feldspar	0.0
Biotite	0.0
Quartz	0.0
Magnetite	0.0
Ilmenite	0.0

Table 6: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
wt.%	40.0	0.25	1.0	17.0	40.0	1.25	0.25	0.25

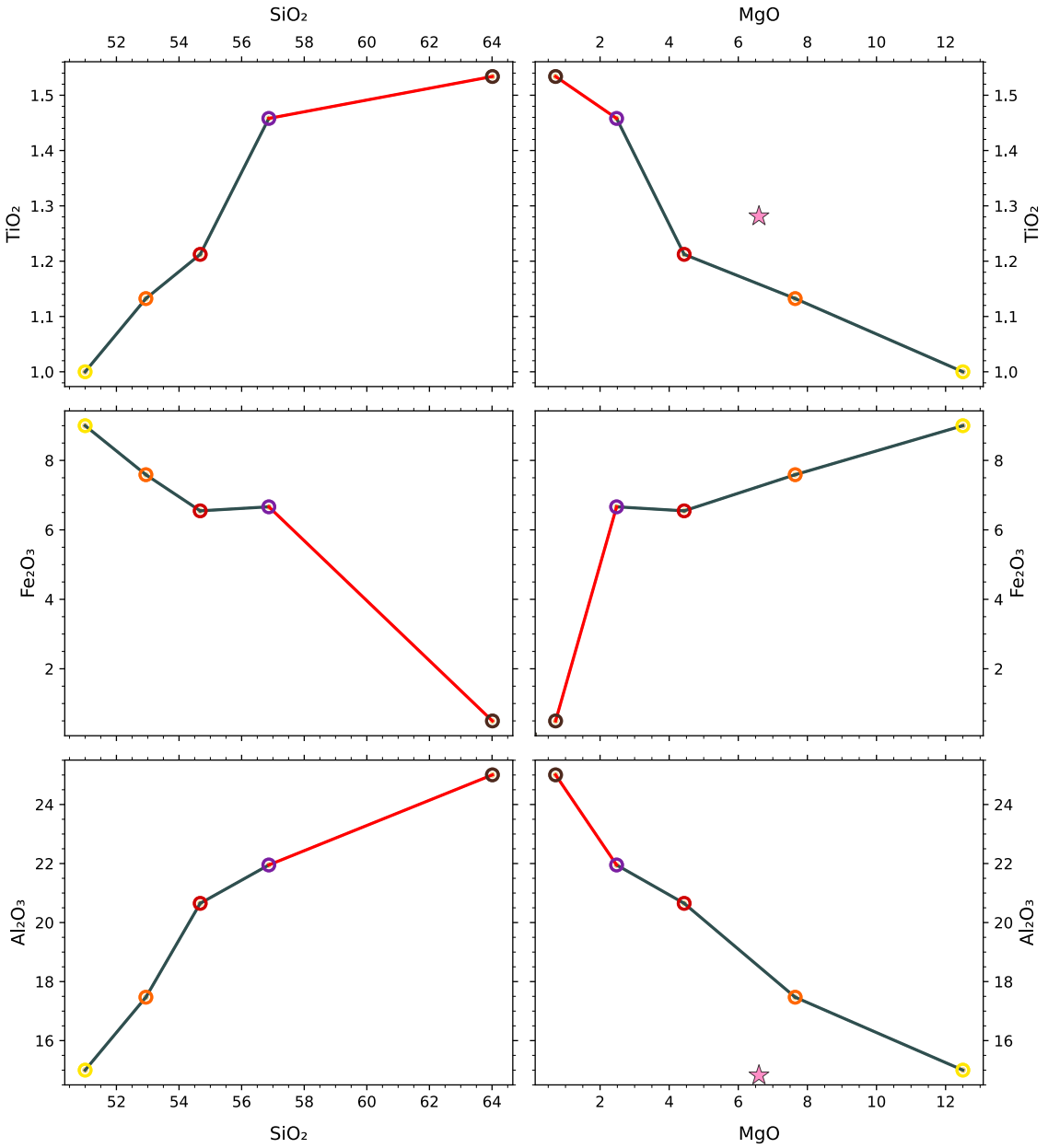
Table 7: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
olivine								
spinel								
clinopyroxene	40.0	0.25	1.0	17.0	40.0	1.25	0.25	0.25
orthopyroxene								
plagioclase								
amphibole								
kfeldspar								
biotite								
quartz								
magnetite								
ilmenite								

Table 8: Cotectic compositions of the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

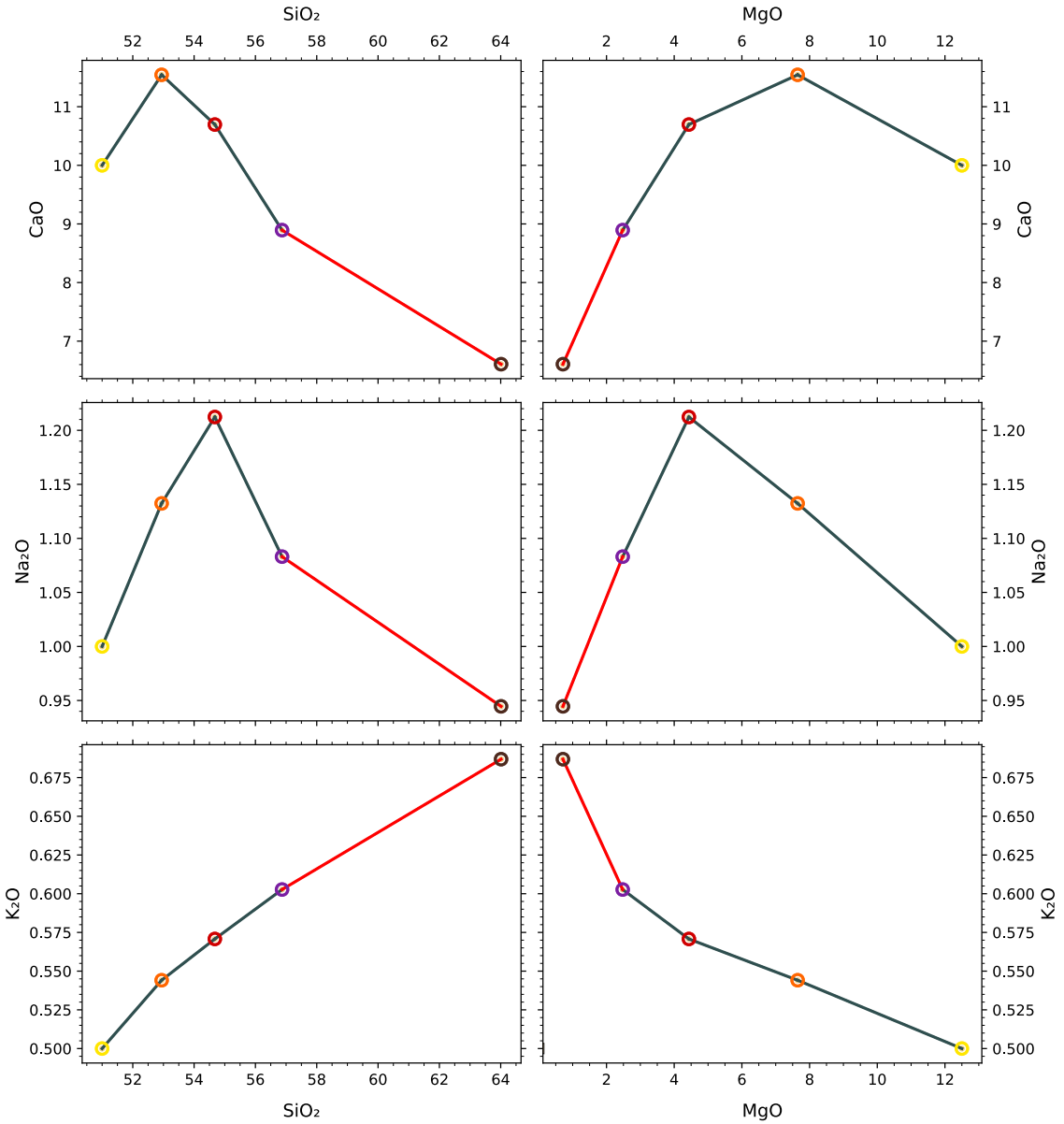


Figure 1: Bivariate plots of the data suite with the data-based and modelled LLODs. The Primary to Primitive transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

Primitive > Basic

PRIMITIVE

Population size	100
Iterations	2394/5000
Cost function	0.00
Trace analysis?	False

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	52.94	54.68	54.68	0.0
TiO2	1.13	1.21	1.21	0.0
Al2O3	17.47	20.65	20.65	0.0

	Initial	Expected	Computed	% Error
Fe2O3	7.59	6.55	6.55	0.0
MgO	7.65	4.43	4.43	0.0
CaO	11.54	10.7	10.7	0.0
Na2O	1.13	1.21	1.21	0.0
K2O	0.54	0.57	0.57	0.0

Table 9: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	20.0
Olivine	0.0
Spinel	0.0
Clinopyroxene	25.0
Orthopyroxene	75.0
Plagioclase	0.0
Amphibole	0.0
K-feldspar	0.0
Biotite	0.0
Quartz	0.0
Magnetite	0.0
Ilmenite	0.0

Table 10: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
wt.%	46.0	0.81	4.75	11.75	20.5	14.94	0.81	0.44

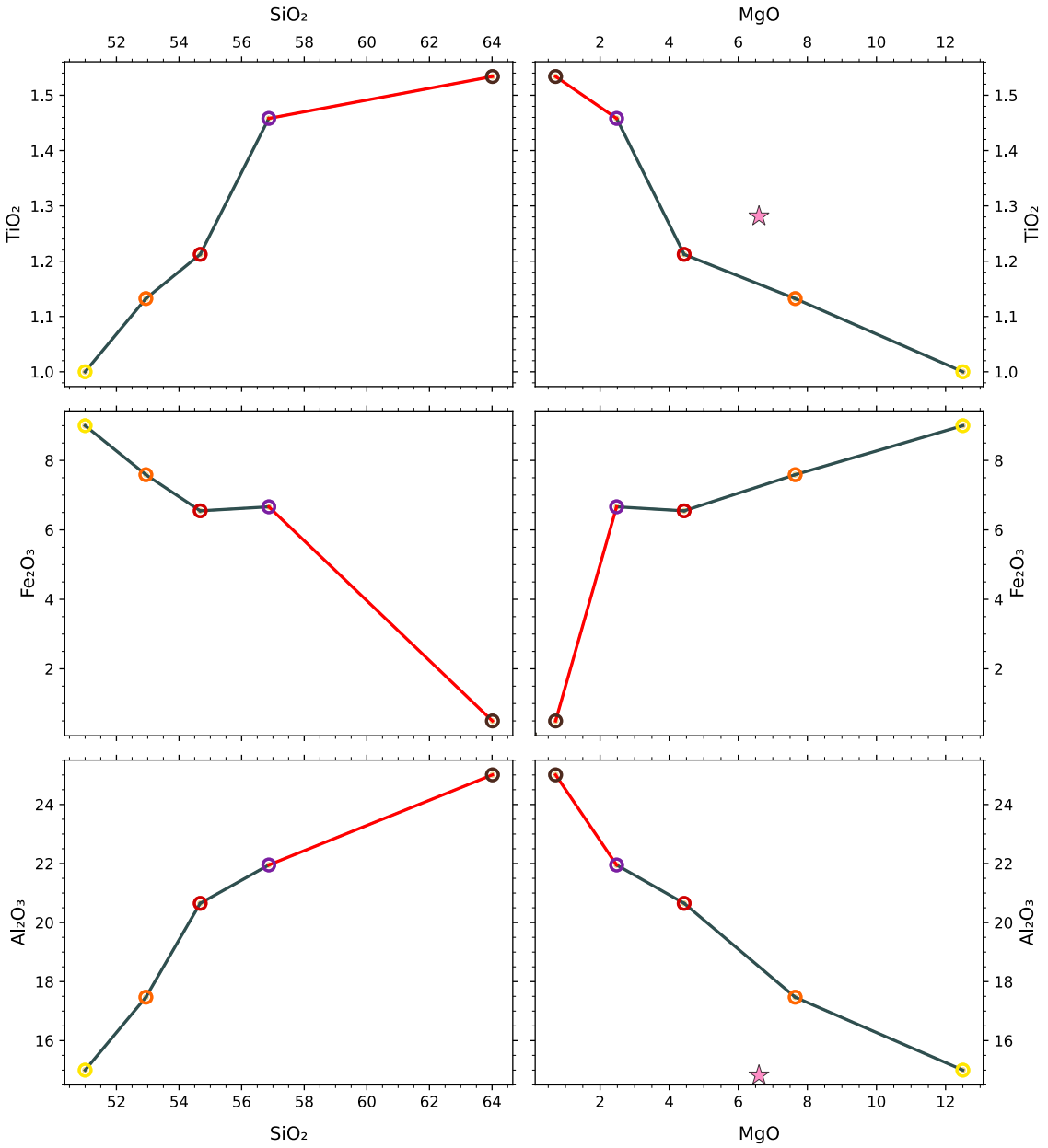
Table 11: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
olivine								
spinel								
clinopyroxene	40.0	0.25	1.0	17.0	40.0	1.25	0.25	0.25
orthopyroxene	48.0	1.0	6.0	10.0	14.0	19.5	1.0	0.5
plagioclase								
amphibole								
kfeldspar								
biotite								
quartz								
magnetite								
ilmenite								

Table 12: Cotectic compositions of the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

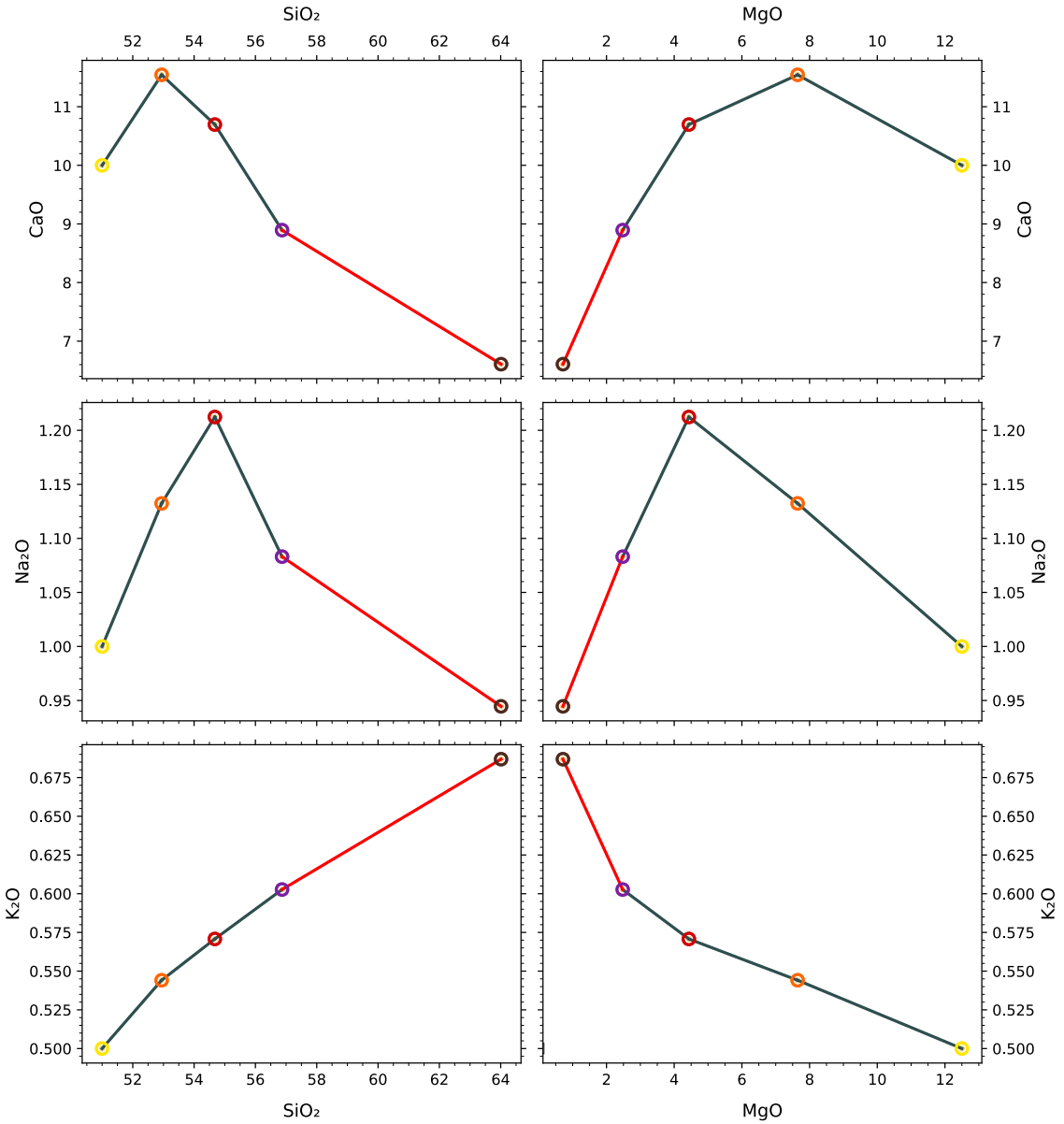


Figure 2: Bivariate plots of the data suite with the data-based and modelled LLODs. The Primitive to Basic transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

Basic > Intermediate

BASIC

Population size	100
Iterations	5000/5000
Cost function	0.00
Trace analysis?	False

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	54.68	56.87	56.87	0.0
TiO2	1.21	1.46	1.46	0.0
Al2O3	20.65	21.95	21.95	0.0

	Initial	Expected	Computed	% Error
Fe2O3	6.55	6.66	6.66	0.0
MgO	4.43	2.48	2.48	0.0
CaO	10.7	8.89	8.89	0.0
Na2O	1.21	1.08	1.08	0.0
K2O	0.57	0.6	0.6	0.0

Table 13: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	25.0
Olivine	0.0
Spinel	0.0
Clinopyroxene	10.0
Orthopyroxene	45.0
Plagioclase	45.0
Amphibole	0.0
K-feldspar	0.0
Biotite	0.0
Quartz	0.0
Magnetite	0.0
Ilmenite	0.0

Table 14: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
wt.%	48.1	0.48	16.75	6.2	10.3	16.1	1.6	0.48

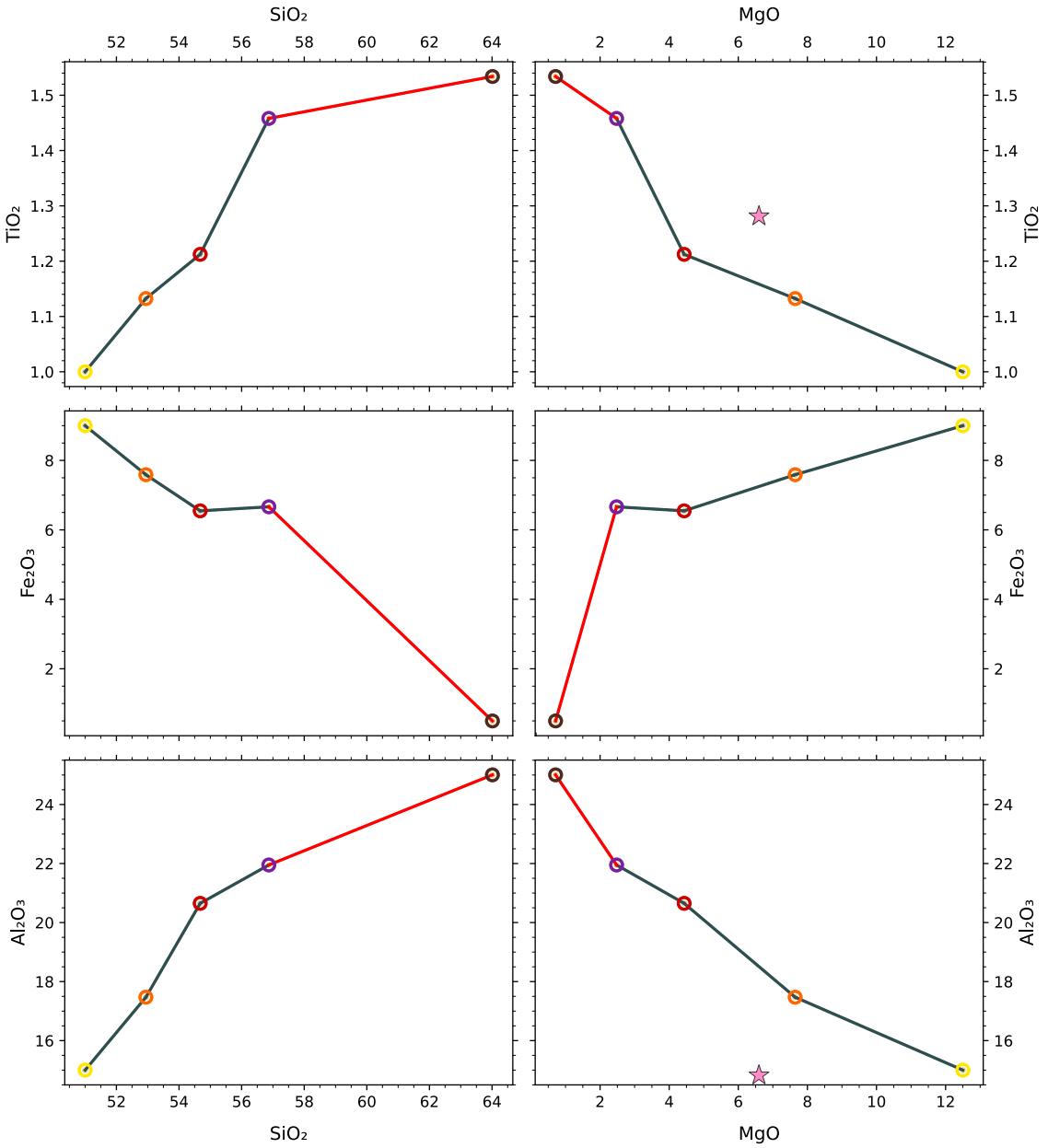
Table 15: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
olivine								
spinel								
clinopyroxene	40.0	0.25	1.0	17.0	40.0	1.25	0.25	0.25
orthopyroxene	48.0	1.0	6.0	10.0	14.0	19.5	1.0	0.5
plagioclase	50.0	0.0	31.0	0.0	0.0	16.0	2.5	0.5
amphibole								
kfeldspar								
biotite								
quartz								
magnetite								
ilmenite								

Table 16: Cotectic compositions of the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

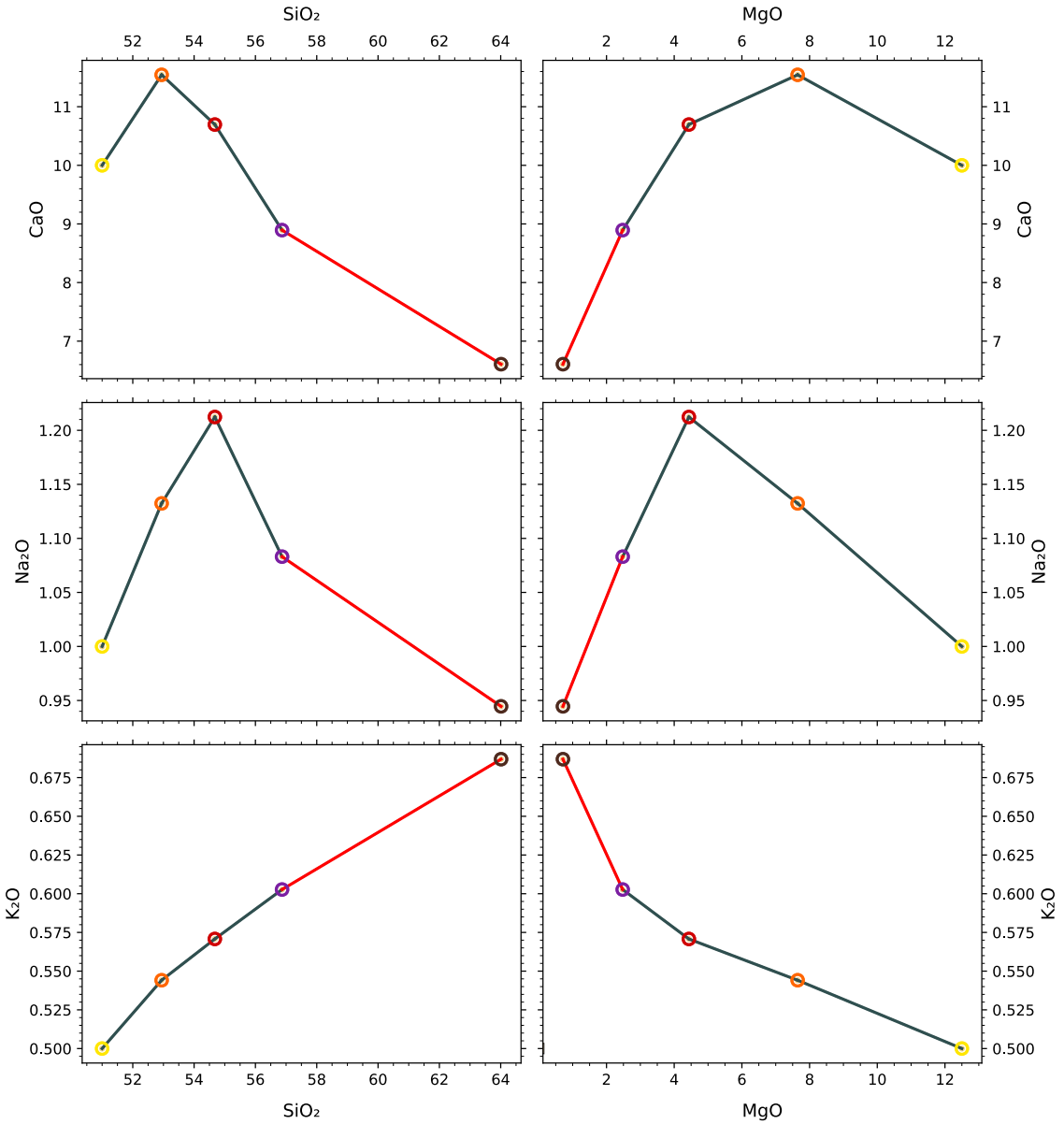


Figure 3: Bivariate plots of the data suite with the data-based and modelled LLODs. The Basic to Intermediate transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

Intermediate > Evolved

INTERMEDIATE

Population size	100
Iterations	5000/5000
Cost function	0.00
Trace analysis?	False

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	56.87	64.01	64.01	0.0
TiO2	1.46	1.53	1.53	0.0
Al2O3	21.95	25.0	25.0	0.0

	Initial	Expected	Computed	% Error
Fe2O3	6.66	0.5	0.5	0.06
MgO	2.48	0.71	0.71	0.09
CaO	8.89	6.61	6.61	0.0
Na2O	1.08	0.94	0.94	0.0
K2O	0.6	0.69	0.69	0.0

Table 17: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	30.0
Olivine	0.0
Spinel	0.0
Clinopyroxene	2.5
Orthopyroxene	40.0
Plagioclase	40.0
Amphibole	17.5
K-feldspar	0.0
Biotite	0.0
Quartz	0.0
Magnetite	0.0
Ilmenite	0.0

Table 18: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
wt.%	40.2	1.28	14.83	21.05	6.6	14.23	1.41	0.41

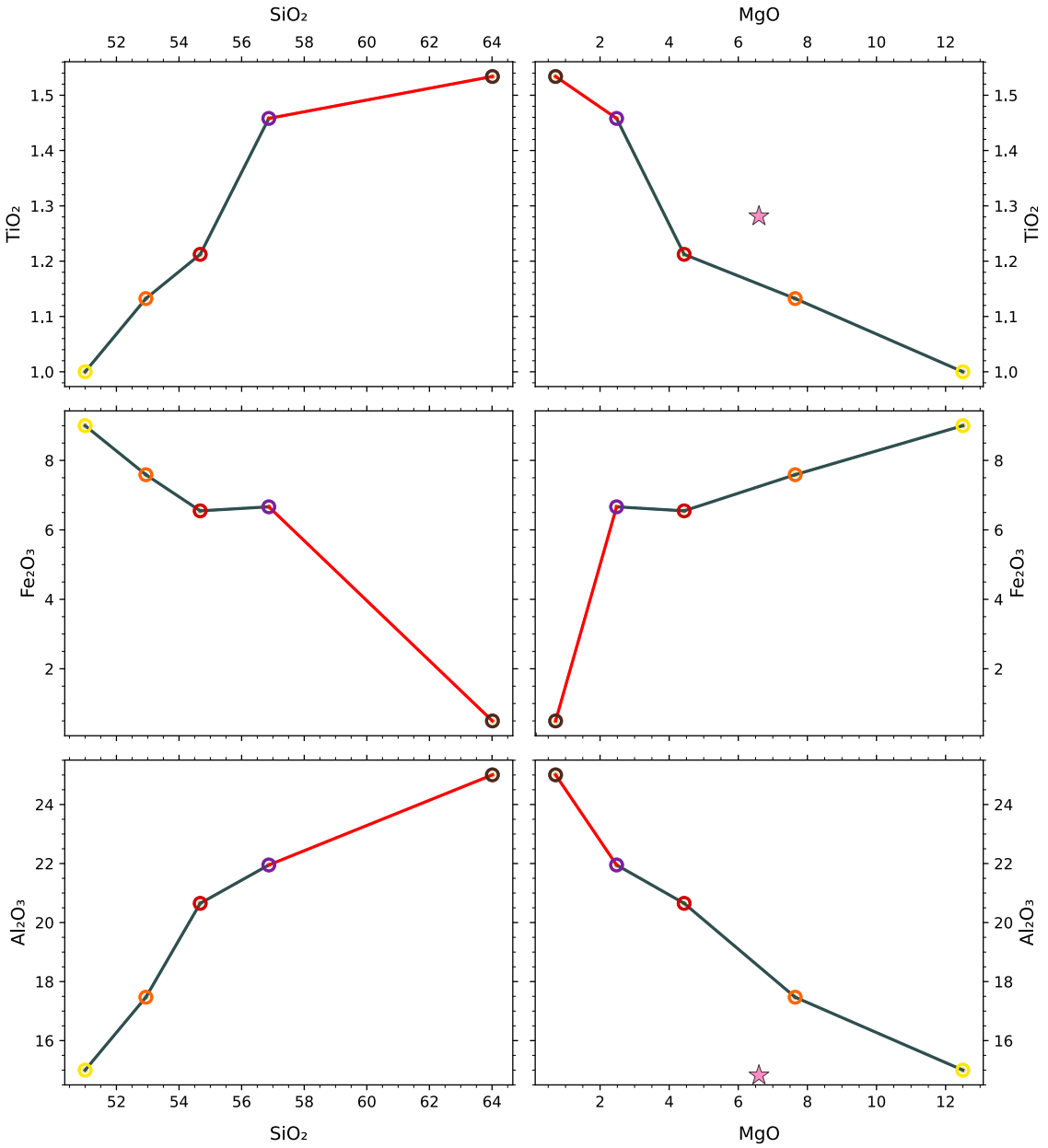
Table 19: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
olivine								
spinel								
clinopyroxene	40.0	0.25	1.0	17.0	40.0	1.25	0.25	0.25
orthopyroxene	48.0	1.0	6.0	10.0	14.0	19.5	1.0	0.5
plagioclase	50.0	0.0	31.0	0.0	0.0	16.0	2.5	0.5
amphibole	0.0	5.0	0.0	95.0	0.0	0.0	0.0	0.0
kfeldspar								
biotite								
quartz								
magnetite								
ilmenite								

Table 20: Cotectic compositions of the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

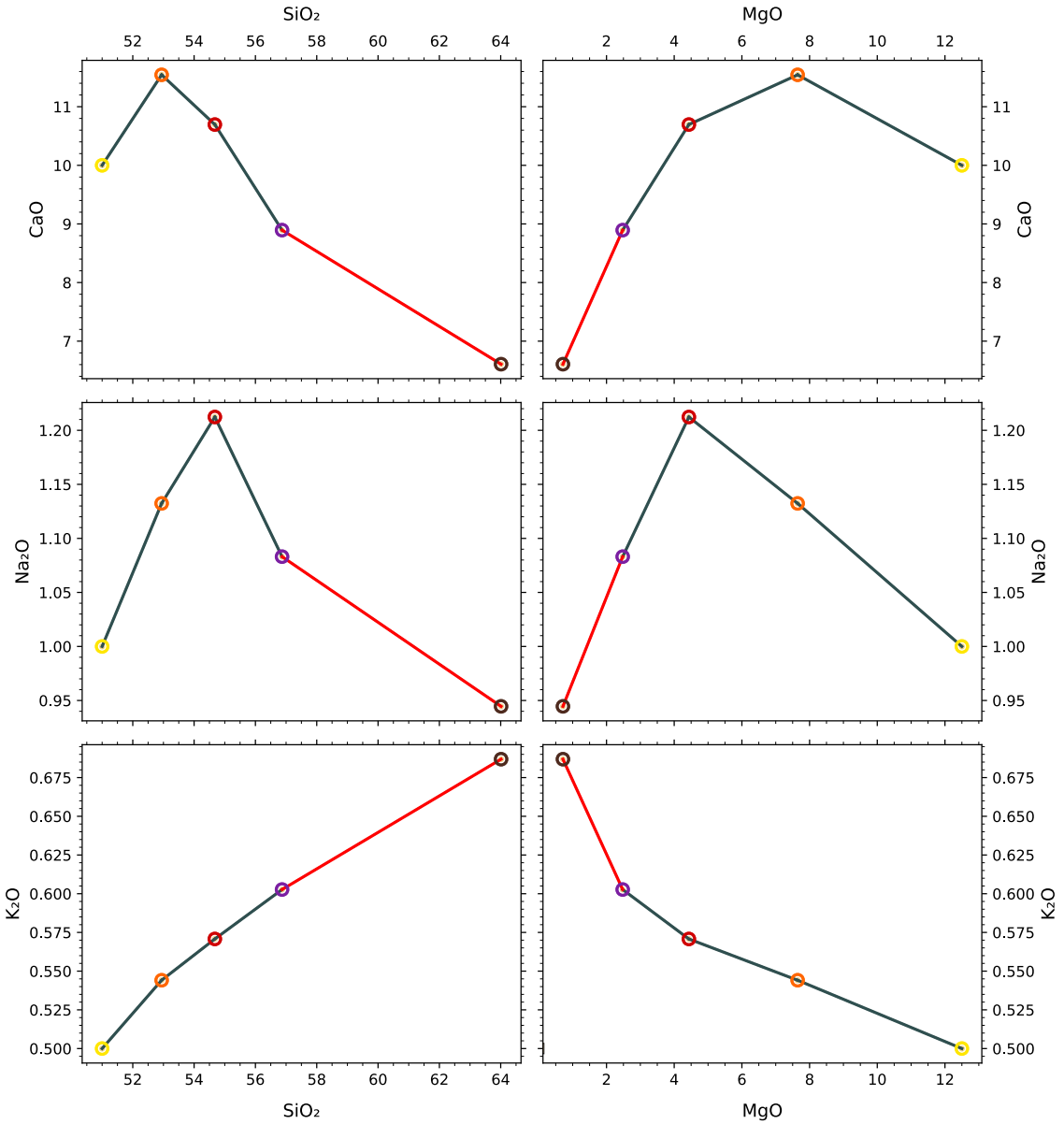


Figure 4: Bivariate plots of the data suite with the data-based and modelled LLODs. The Intermediate to Evolved transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

5. Figures

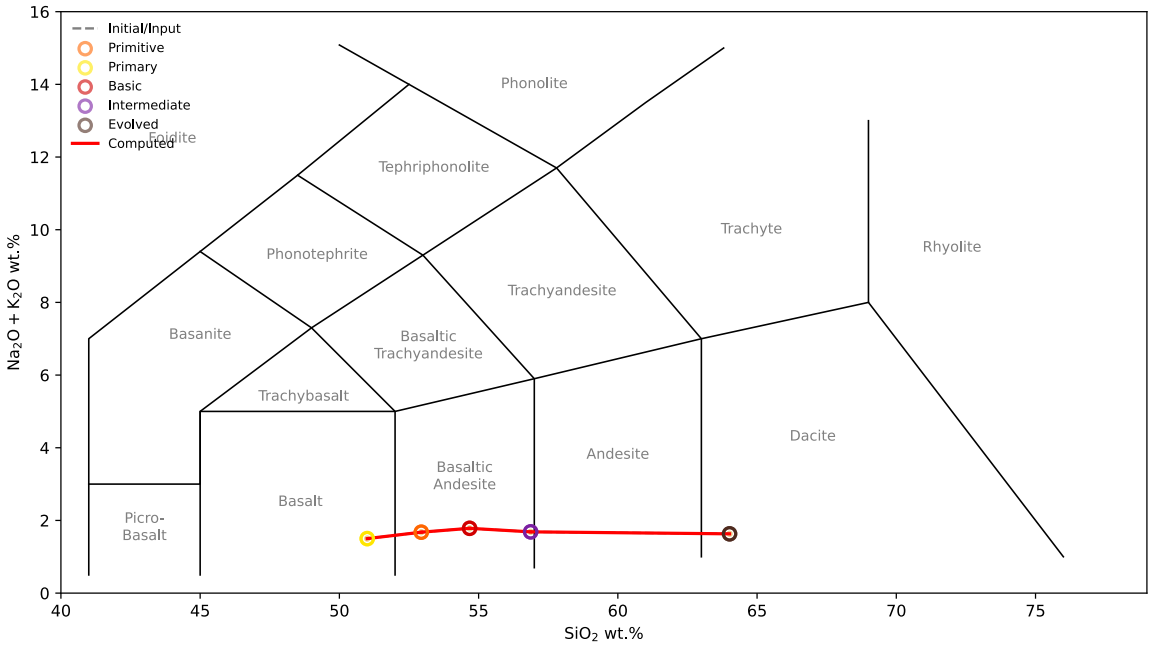


Figure 5: Total Alkali-Silica (TAS) diagram of the model phases.

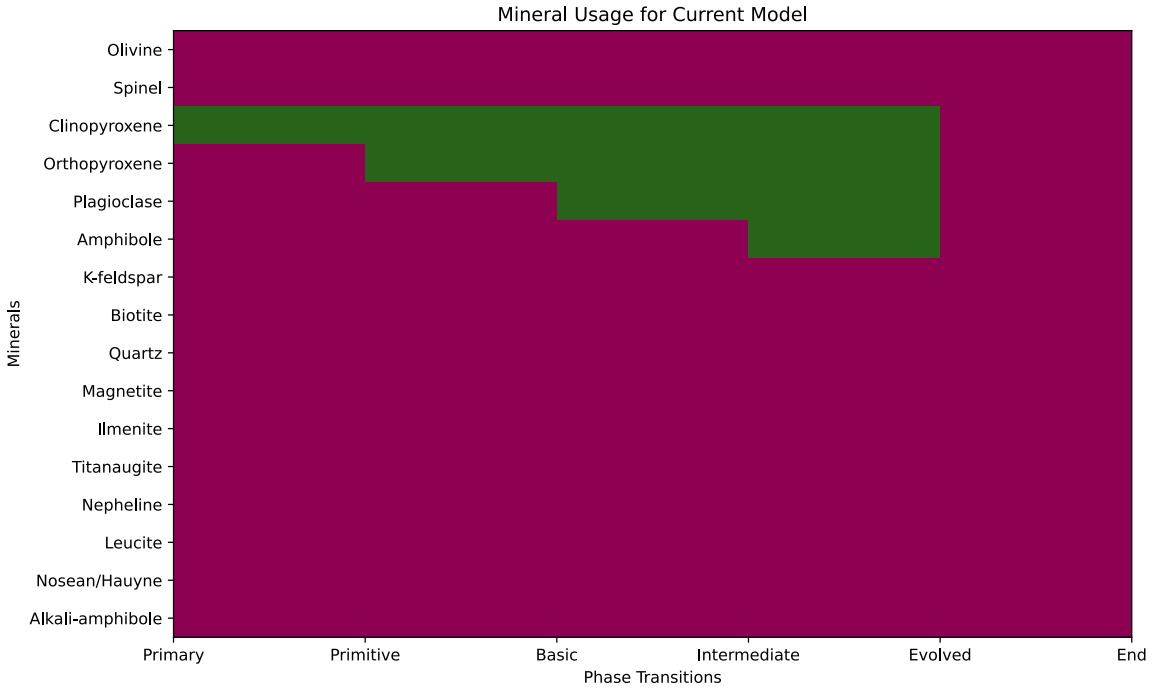


Figure 6: Mineral usage across the model phases.

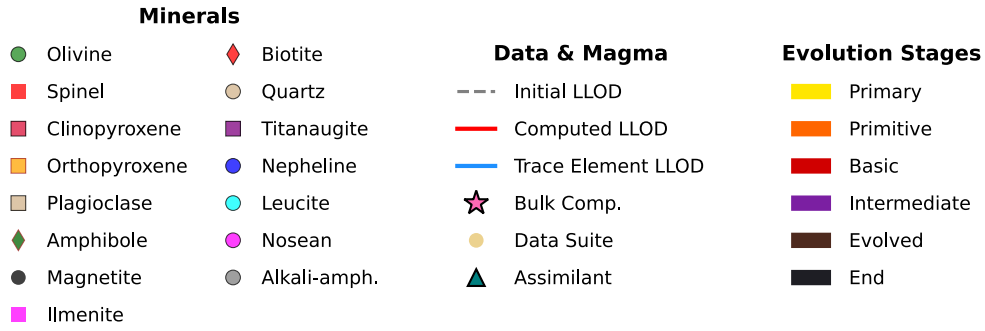


Figure 7: Legend for mineral symbols used in the figures.